Earth and Planetary Science Letters, 30 (1976) 269–280
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PALAEOMAGNETIC AND GEOCHRONOLOGICAL WORK IN CENTRAL WEST GREENLAND

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Received November 13, 1975
Revised version received February 26, 1976

A palaeomagnetic pole at 56.9°N, 273.6°E (dp = 13.3°, dm = 13.8°) has been determined from dykes and gneisses from Sagdlerssuaq and adjacent islets; this magnetization was probably acquired at a time given by a K–Ar age of 1620 (±50) Ma. Dykes at Itivdleq have given a palaeomagnetic pole at 18.0°N, 287.2°E (dp = 5.4°, dm = 7.2°); these dykes were probably magnetized at about 1750 Ma. It seems likely that the magnetizations were imposed upon cooling during uplift following the Nagssugtoqidian metamorphism.

1. Introduction

This work was undertaken in the hope [1] of providing information on the thermal history of the Nagssugtoqidian boundary near Holsteinsborg (Fig. 1). The Nagssugtoqidian metamorphism took place in early Proterozoic times. Distinct phases have now been distinguished within this metamorphism [1,2]. The main phase known as Nag 1 produced shear zones such as the one along Itivdleq fjord. Then two sets of dolerite dykes were emplaced, a sparse east–west swarm followed by the extensive Kangâmiut dykes. The Kangâmiut dykes strike NNE in the Archaean block but run ENE in regions affected by Nag 1. This effect is due to the dykes having followed lines of weakness in the shear zones. In spite of the change of direction the dykes are mainly undeformed and unmetamorphosed in the Itivdleq shear zone [2].

About 20 km to the north of Itivdleq (typically in Sagdlerssuaq) a later Nagssugtoqidian event, Nag 2, has been recognized which deformed and metamorphosed the dykes into strongly sheared amphibolites. Both these phases took place whilst the rocks were still at high temperatures and ductile [2]. The history of the Sagdlerssuaq area is complicated by ductile overthrusting but this was absent in Itivdleq fjord [3]. Brittle movements (often with pseudotachylite formation) occurred extensively in the whole region from

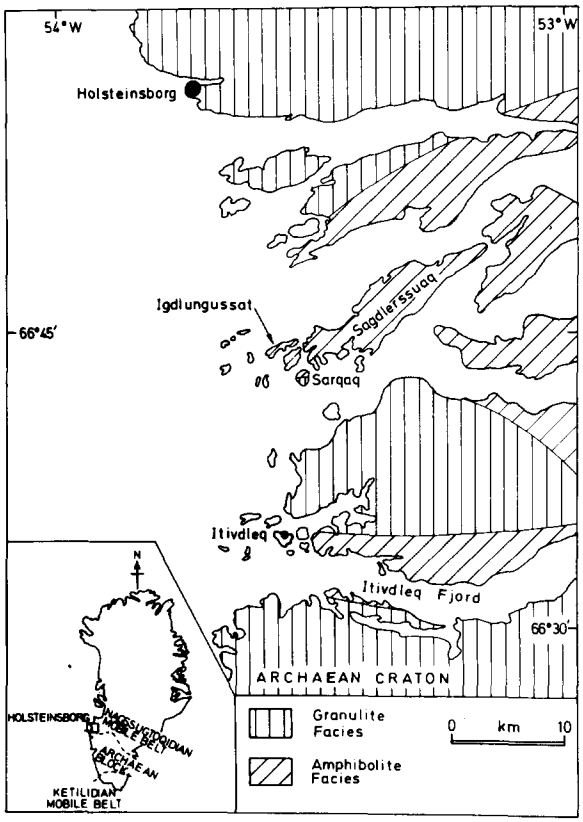


Fig. 1. The sampling area showing metamorphic facies distribution (after Bak et al. [3]).

the end of the Nagssugtoqidian metamorphism until Mesozoic times [4]. Another aspect of this activity was the intrusion of a suite of kimberlite dykes in the Cambrian (Bridgwater [5] and R.M. Macintyre, personal communication, 1976).

The rocks collected for this study consisted of gneisses and the Kangâmiut and kimberlite dykes. The palaeomagnetic work is concerned mainly with material from Sagdlerssuaq and two adjacent islets but includes a few samples from Itivdleq. The geochronological work was devoted equally to both areas.

2. Palaeomagnetic work

2.1. Sampling and laboratory procedures

Sampling was carried out using a portable rock-drill; cores were drilled approximately 7 cm long and $2\frac{1}{2}$ cm diameter. Most of the orientation was done with the sun-compass designed by Stone [6], and the remainder using a Brunton compass in conjunction with topographic sighting. After the field work a specimen about $2\frac{1}{2}$ cm long was cut from each sample. Measurements of the intensity and direction of magnetization were all made on the PAR SM-2 spinner magnetometer. Magnetic cleaning was effected mainly by progressive alternating field (AF) demagnetization, whilst specimens tumbled about three axes, with the Earth's field annulled to an accuracy of ±20 gammas. A limited amount of cleaning by thermal demagnetization has also been done.

2.2. Sagdlerssuaq

The island of Sagdlerssuaq consists of strongly foliated acid and basic gneisses formed in amphibolite facies by Nag 2. The sampling was concentrated on five basic bands which may originally have been Kangâmiut dykes; these are generally about 10 cm wide but some of them have widths of a few metres. The specimens from four of these bands yielded very well-grouped natural remanent magnetization directions but the fifth was less well-grouped. The directions for these bands fall close to each other (Fig. 2a). They also lie near the Earth's present field direction (inclination = 81°, declination = 313°) a fact which prompted an extensive series of demagnetizing experiments. The intensities fall sharply for relatively small demagnetizing fields (Fig. 3a). However, the changes of mean direction are small, often amounting only to 1° or 2° at each stage (Fig. 2a). It is believed that this behaviour is due to two components of magnetization which happen to have nearly the same direction, namely a secondary component, perhaps due to the present field, superimposed upon an ancient magnetization. The cleaned directions (Table 1) have been taken to be those for which the *mean* magnetic vector has moved least in a demagnetizing step. At this stage the precision factor k of each band is near but not necessarily at its peak value (because the initial grouping is sometimes dominated by the secondary component).

The AF work has been checked by demagnetizing nine specimens thermally in steps in an air furnace operating in field-free space. The mean directions are close to those established by the alternating field method (Fig. 2c). Particularly stable behaviour occurred for three specimens from band G8 for which the mean direction altered only 2° between room temperature and 550°C whilst the intensities fell to about 10% of the original values; the individual directions are so close that they cannot be shown as distinct points on a stereogram.

Three kimberlite dykes were also sampled on Sagdlerssuaq. The mean directions after cleaning are very widely separated. These results might reflect different ages within the Cambrian which may have been a period of very rapid polar wander [8] (although the latter suggestion does not seem to be substantiated by Russian data [9, pp. 296–297]).

2.3. Sarqaq

Off the southern tip of Sagdlerssuaq lies the islet of Sarqaq through which the Nag 2 boundary passes. A few metres to the south of this boundary there is an undeformed Kangâmiut dyke displaying orthopyroxene weathering and original igneous texture. The 17 specimens collected from this dyke were each subjected to seven stages of AF demagnetization (Fig. 2b). The directions are well-grouped up to and including the 350-Oe stage; however, they are nearly random after 500 Oe. The rate of decline of intensity is similar to that of the Sagdlerssuaq gneisses, implying that the magnetic material has comparable coercivities. Demag-

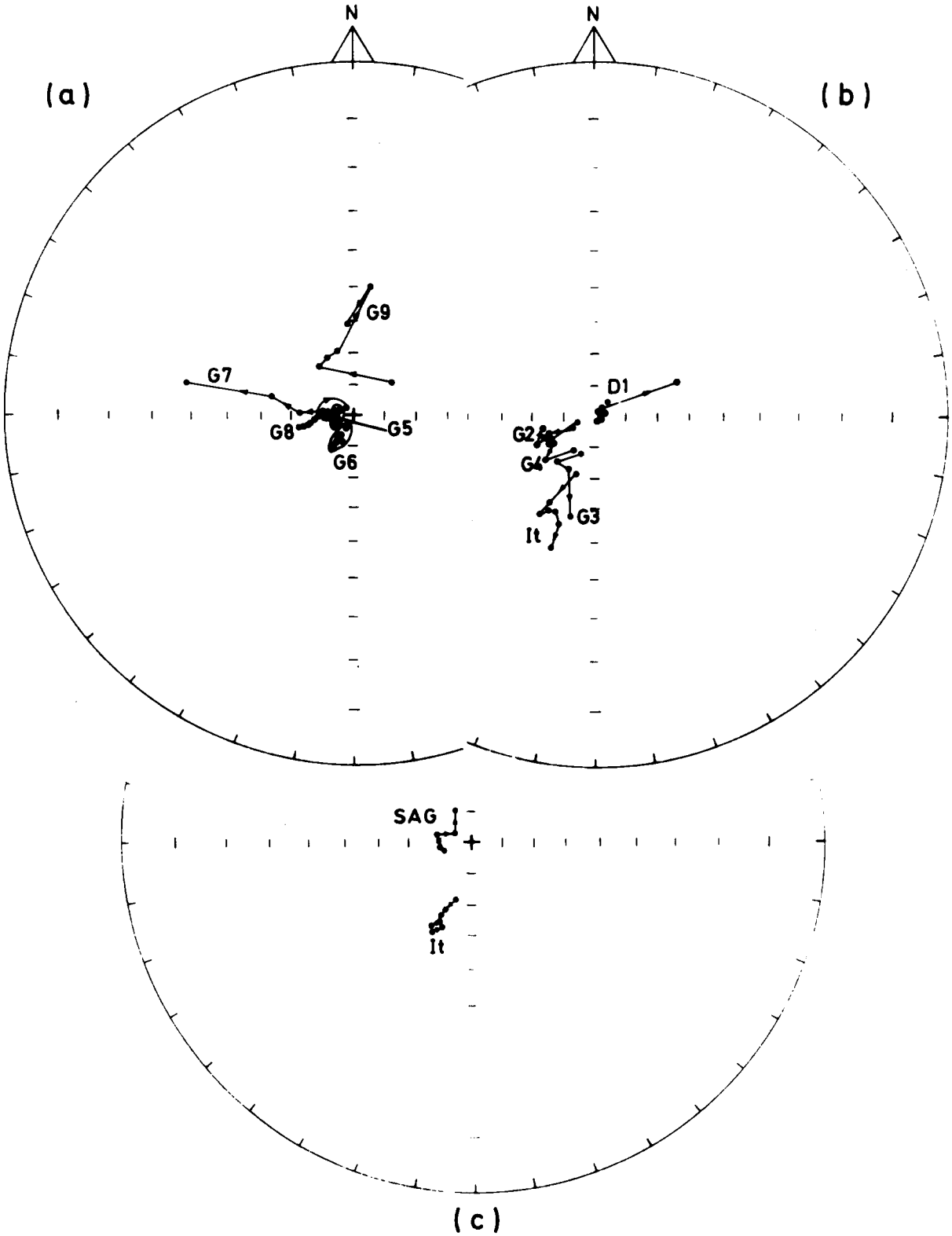


Fig. 2. (a) Mean directions of magnetization of the Sagdlerssuaq gneiss bands after AF demagnetization at 25, 50, 100, 150, 250 Oe (peak). (b) Mean directions after AF demagnetization: Sarqaq dyke, D1, after 0, 25, 50, 100, 150, 250, 350, 500 Oe; Igdlungussat units G2 and G4 after 0, 25, 50, 100, 150, 250 Oe and G3 after 0, 25, 50, 100 Oe; Itivdleq dykes (It) after 0, 100, 200, 300, 500, 700, 900 Oe. (c) Mean directions after thermal demagnetization: It = 17 specimens of Itivdleq dykes after 20, 200, 300, 400, 500, 540, 560°C. SAG = 9 specimens of the gneiss bands subjected to similar temperatures.

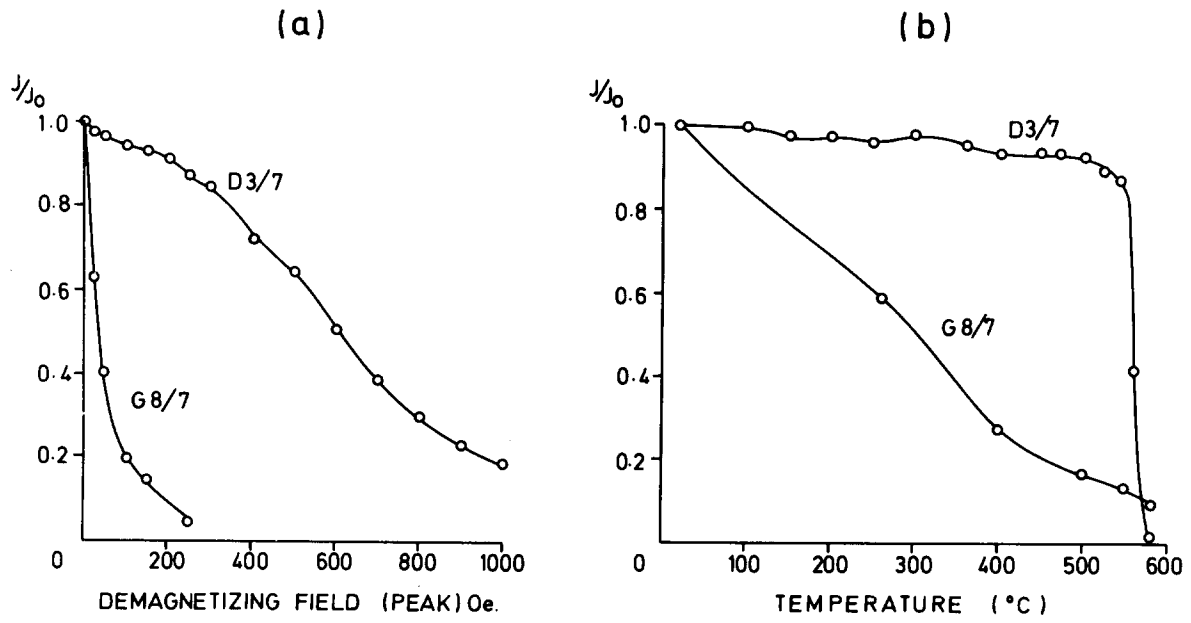


Fig. 3. (a) Comparison of intensities after AF demagnetization for an Itivdleq dyke specimen (D3/7) and a Sagdlerssuaq gneiss specimen (G8/7). (b) Comparison of intensities after thermal demagnetization for specimens cut from the same samples.

netization causes only small changes in mean direction until the 500-Oe stage. Moreover these directions fall close to those for the Sagdlerssuaq gneisses, in spite of the fact that the dyke is undeformed whereas the gneisses were strongly deformed by Nag 2; this suggests that the magnetizations were imposed after that event. Apparently the foliation in the gneisses is not an important factor influencing their direction of magnetization. (Indeed it has been demonstrated [10] that there is no objection to using "visibly foliated gneissic rocks" for the calculation of palaeomagnetic poles.)

2.4. *Igdlungussat*

On this islet, 2 km west of Sagdlerssuaq, three rock units were sampled, two of which were recognizable as deformed Kangâmiut dykes, the third being a gneissified volcanic in the form of a rust band. The two dykes (G2 and G4) responded very similarly to AF demagnetization which was performed at five stages for each specimen. The directions of magnetization are almost constant at the 100-Oe stage (Fig. 2b). On the other hand, the rust band (G3) is poor palaeomagnetic material in that demagnetization at small fields caused scattering of the directions; after the 100-Oe stage the grouping is nearly random and so no further demagnetization was carried out.

2.5. *Itivdleq*

The Kangâmiut dykes at Itivdleq are often very stably magnetized [11,12] and so it seemed of interest to make a detailed comparison of the AF and thermal demagnetization characteristics of a few specimens. Accordingly 17 samples from five dykes were cut into two specimens each; one was given AF and the other thermal treatment. The remanence vanishes at about 580°C (Fig. 3b) demonstrating that it is borne by magnetite. For both investigations the directions are well-grouped and the mean direction changes very little at each step ("It" in Figs. 2b and 2c); moreover the optimum mean directions are almost identical (Table 1 and Fig. 4). Also the high value of the precision parameter (bearing in mind that the specimens are drawn from five different dykes) suggests that secular variation is not present. Such behaviour would be expected if each specimen has been magnetized over a very long period.

A kimberlite was also sampled at Itivdleq and has

TABLE 1

Palaeomagnetic sampling and results

Locality	Rock type	Site codes		Number of specimens	Optimum demag. field (peak Oe)	Magnetization *					Virtual geomagnetic pole			
						inc.	dec.	R	k	α_{95}	lat.	long.	dp	dm
Sagdlerssuaq	basic gneiss bands	G5		12	50	85.0	263.5	11.94478	199	3.1	63.8°N	283.8°E	6.0	6.1
		G6		11	50	81.0	211.0	10.92578	135	3.9	50.7°N	292.3°E	7.4	7.6
		G7		13	50	81.3	277.5	12.92684	164	3.2	63.3°N	266.2°E	6.1	6.3
		G8		13	25	72.9	256.9	12.94394	214	2.8	47.3°N	257.6°E	4.5	5.1
		G9		5 **	50	69.9	337.9	4.67991	12	22.5	73.3°N	176.9°E	33.2	38.7
	kimberlite dykes	K1		9	50	12.0	268.2	7.89669	7	20.5	4.9°N	220.6°E	10.6	20.9
		K2		7	100	47.4	291.3	6.62817	16	15.5	34.4°N	209.2°E	13.0	20.1
		K2£		3 ***	150	80.7	161.4	2.99719	711	4.6	49.2°N	315.2°E	8.6	8.9
Sarqaq	undeformed Kangâmiut dyke	D1		17	150	88.6	114.9	16.76162	67	4.4	65.4°N	312.7°E	8.8	8.8
Igdlungussat	deformed Kangâmiut dykes	G2		13	100	74.2	249.6	12.80924	63	5.3	47.0°N	263.9°E	8.6	9.5
		G4		11 **	100	73.3	240.0	10.60257	25	9.3	43.3°N	268.8°E	14.9	16.6
	volcanic gneiss (rust band)	G3		12	50	70.8	207.0	9.83137	5	21.4	33.6°N	288.4°E	32.4	37.2
Itivdleq	kimberlite dyke	K3		9	150	14.5	268.1	7.68490	6	22.7	6.0°N	221.1°E	11.9	23.3
	undeformed Kangâmiut dykes	D2 to D7 D14, D15 D17	AF	17	200	57.1	206.6	16.68939	52	5.0	16.2°N	284.9°E	5.3	7.3
			thermal	17	540(°C)	59.1	204.4	16.71501	56	4.8	18.0°N	287.2°E	5.4	7.2

* The declination of the mean direction of magnetization is measured in degrees east of true north and its inclination from the horizontal and regarded as positive downwards. The statistics are those of Fisher [7]; k is a measure of the precision of grouping of directions and α_{95} is the semiangle of the cone of 95% confidence about the mean direction. The virtual geomagnetic pole is calculated from the mean direction; dp and dm are the semiaxes of the oval of confidence about this pole at the 95% probability level, dp in the colatitude direction and dm perpendicular to it.

** There were 7 further specimens in G9 and 2 in G4 which were too weakly magnetized to measure.

*** These 3 specimens occurred in a dyke which was believed in the field to be a branch of K2 but the palaeomagnetic results suggest that it was not.

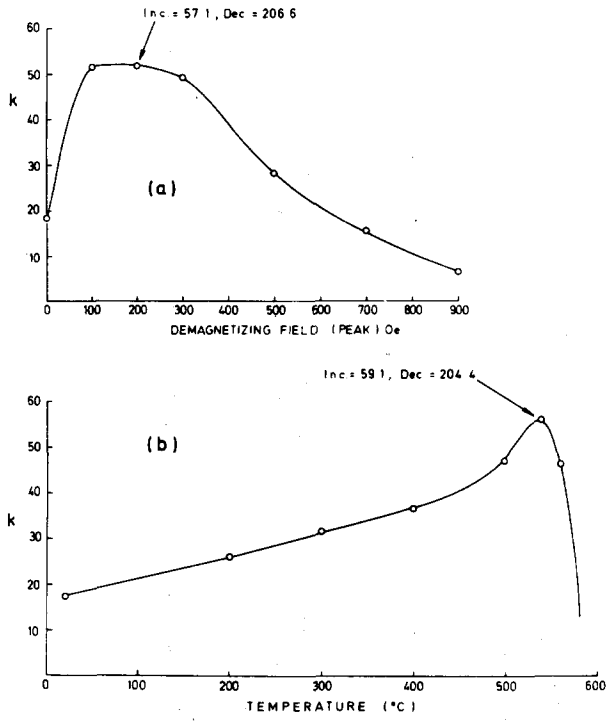


Fig. 4. Comparison of the behaviour of the precision parameter for the directions of magnetization of the 17 Itivdleq dyke specimens (from 5 dykes) upon (a) AF, (b) thermal demagnetization.

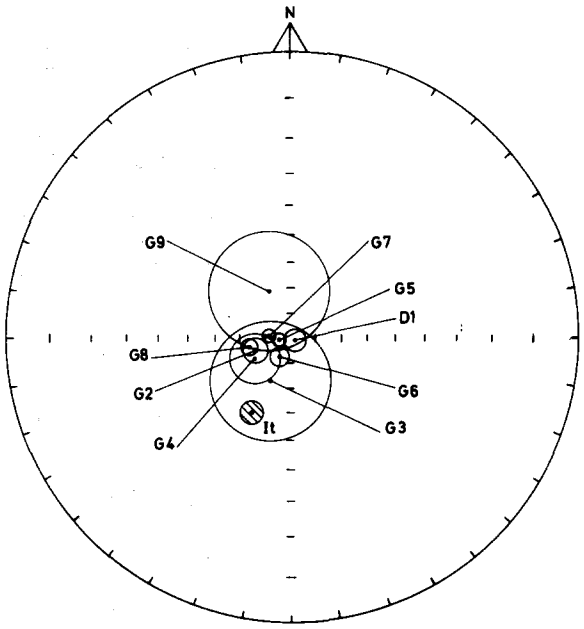


Fig. 5. Mean directions of magnetization surrounded by their circles of 95% confidence on equal area projection, for all the Precambrian units of the study (illustrating data in Table 1). All these rocks are normally magnetized.

given the same mean direction of magnetization as one of those from Sagdlerssuaq but the errors are large. The palaeomagnetic poles from the four kimberlites in this survey are unlike Cambrian poles from North America [9,284].

2.6. Assessment of the palaeomagnetic work

The optimum directions for all the Precambrian rocks after AF demagnetization are shown in Fig. 5. The rock units from Sagdlerssuaq (apart from the kimberlites) and the two adjacent islets give directions sufficiently close to justify forming an overall result (Table 2) for the Sagdlerssuaq region, assuming a Fisherian distribution for the optimum mean directions for each site. The proximity of the direction of magnetization at Sagdlerssuaq to that of the Earth's present field implies that the remanence could be of recent origin, perhaps a chemical remanent magnetization (CRM) due to weathering. Such a low-temperature oxidation phase in magnetite might be expected to invert upon heating. However, the thermal demagnetization curve (Fig. 3b) for G8/7, which has been obtained by repeated heating and cooling, shows no sign of this; it is continuous and the changes of direction of the magnetic vector are always less then $2\frac{1}{2}^\circ$ between each point.

Experiments by Johnson and Merrill [13] have shown that when single-domain magnetite is completely oxidized to maghemite at 50°C, no CRM was acquired in the field applied during oxidation; on the contrary, the original direction of magnetization persisted. These authors have suggested that even for larger grains of magnetite, low-temperature oxidation may be acquired over a time sufficiently long for any CRM to be randomized on account of field reversals leaving only the original remanence.

These considerations, combined with the evidence from AF demagnetization, favour the conclusion that the direction of magnetization at Sagdlerssuaq is a

TABLE 2

Mean palaeomagnetic result for the Sagdlerssuaq region (excluding the kimberlites)

Number of sites	Magnetization					Virtual geomagnetic pole			
	inc.	dec.	R	k	α_{95}	lat.	long.	dp	dm
9	80.7	253.1	8.84699	52	7.2	56.9°N	273.6°E	13.3	13.8

genuine Precambrian direction. The corresponding palaeomagnetic pole lies in northeast North America (Fig. 6) and so is well to the north of that from the Itivdleq region. Our Itivdleq pole rests upon 17 specimens only, and is quoted mainly for comparison with the Sagdlerssuaq pole. It lies extremely close to those derived by other workers [11,12] in more detailed studies in Itivdleq fjord.

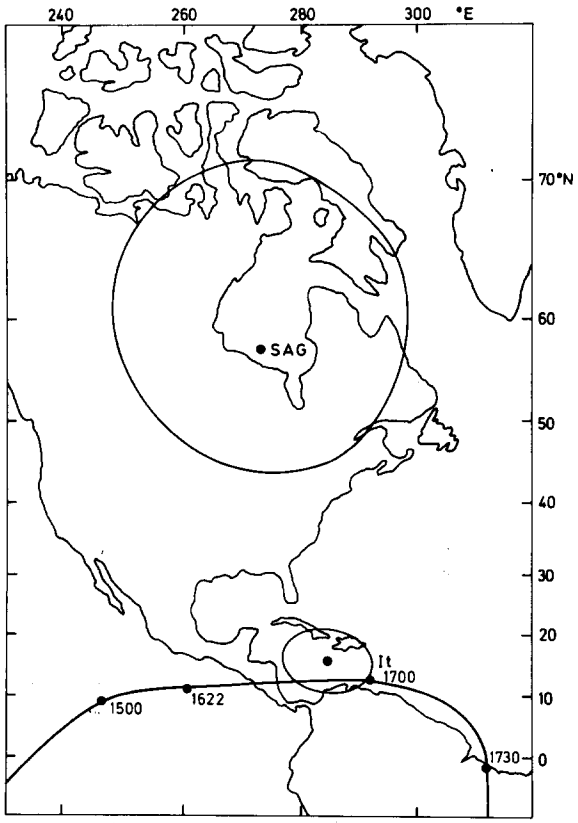


Fig. 6. Palaeomagnetic poles for Sagdlerssuaq (SAG) and Itivdleq (It). Part of pole track 4A [23] for the Canadian Shield is also shown.

3. K–Ar isotopic ages

Previous studies of the Nagssugtoqidian mobile belt have yielded K–Ar ages which ranged from 1790 to 1650 Ma [14]. These ages have been interpreted as expressing the age at which the mineral system in question, biotite, last closed with respect to argon diffusion, although the main phase of the Nagssugtoqidian deformation and metamorphism probably took place near the beginning of the Proterozoic [15]. The argon "blocking temperature" for biotite is probably in the region of 200–300°C [16, p. 157] and accordingly the interpretation of these ages would be that cooling of the exposed structural levels of the mobile belt to temperatures within this range, must have occurred fairly uniformly during a period of the order of 100 Ma. Neither the precision of the analytical data (typically ±40 Ma) nor the density of sampling admits any more detailed interpretation.

K–Ar studies of the rocks described in the present work were undertaken since none of the previous collecting sites lay within 150 km of the sampling area used for palaeomagnetic studies. As is seen from the data we obtained, not only were conventional age determinations possible, but the mineral ages admitted the construction of isochron plots allowing the effects of inherited (metamorphic) ^{40}Ar [16, p. 122] to be assessed and accounted for, leading to a considerable reduction in the spread of K–Ar ages obtained from the suites of rocks analysed. We endorse the view that K–Ar ages obtained from such high-grade metamorphic terrains can express only the time at which the mineral systems in question last closed with respect to argon diffusion, and can have no value in elucidating the earlier history of the mobile belt.

Analyses were performed on gneiss samples from each of three areas, Sagdlerssuaq, Itivdleq and the south coast of Itivdleq Fjord, spanning the area studied by

palaeomagnetic methods. Dyke samples from the latter two areas were also analyzed. Reconnaissance whole-rock analyses (W) were performed on each sample followed by separation of felspathic (L) and mafic (H) minerals (see Table 3) in the case of the gneisses and K–Ar age determinations on the separated fractions to investigate the possible presence of inherited ^{40}Ar. Mineral separation was achieved using bromoform (s.g. 2.98). Potassium was determined by flame photometry and argon by isotope dilution mass spectrometry using ^{38}Ar tracer, on an Omegatron mass spectrometer. The methods have been described elsewhere [17] and the whole-rock ages obtained, together with the mineral ages are presented in Table 3. The data from the gneiss samples is presented in graphical form as isochrons in Fig. 7a and b. We have investigated the data using isotope-ratio plots such as those proposed by Hayatsu and Carmichael [18] and also using radiogenic argon versus potassium plots such as have been used with success by Welin and Lundqvist [19]. The use of such methods in a situation in which each potential "isochron" comprises, at most, three points, must of necessity be tenuous, but the correlations which clearly exist when the latter method of data presentation is employed, we believe does justify its application. No similar regularities exist when the former method of presentation is employed. Fig. 7a and b displays the K–Ar data in graphical form using the latter method and the approximately linear relationships which appear to exist within a single whole-rock sample are consistent with a model process in which a fixed amount of inherited ^{40}Ar is uniformly distributed throughout all the minerals of each analysed sample, rather than with a model in which there exists a non-radiogenic argon fraction whose $^{40}Ar/^{36}Ar$ is uniform throughout each sample. Adopting this model for the interpretation of the K–Ar data, we have calculated the mean age appropriate to each data set, with its associated error, and presented it in Table 3.

Interpreting the data in this way, therefore, the following cooling histories may be established in the sampled areas.

3.1. Sagdlerssuaq

Four of the five "ages" obtained in this manner from the gneisses are concordant with a mean of 1620 ± 50 Ma. The levels of inherited ^{40}Ar range from zero to a maximum value of 4.76×10^{-2} mm^3 g^{-1} in sample G8. The individual mineral ages are discordant due to the incorporation of inherited ^{40}Ar in the cooling mineral systems, the volume distribution of this ^{40}Ar being apparently uniform over the dimension of the sample analysed (~25 mm), and even between samples taken from the same band (cf. G5 and G5/6). The time of closure of the K–Ar system with respect to argon diffusion was, in the Sagdlerssuaq region, 1620 ± 50 Ma, if the postulated model is a realistic one.

3.2. Itivdleq

Of the two gneiss samples analysed, cooling ages of 1700 ± 40 Ma and 1830 ± 20 Ma were obtained from the Ar versus K plots, with inherited ^{40}Ar present unambiguously in one of the samples (G18/11) at a concentration of 5.94×10^{-2} mm^3 g^{-1}. Whole-rock dolerite dyke ages from the area also lie in the range 1720 ± 20 Ma to 1890 ± 20 Ma and strongly suggest that the cooling age of that section of the mobile belt exposed in this area may be significantly older by at least of the order of 100 Ma, relative to the section exposed at Sagdlerssuaq.

3.3. South coast of Itivdleq fjord (Archaean craton)

Three sections of a single gneiss core sample (G11) were analysed and define a line equivalent to an age of 2410 ± 40 Ma with an intercept which is negative and may be an expression of minor argon loss from a mineral assemblage which initially contained an indeterminate amount of inherited ^{40}Ar. We consider that the age obtained probably represents the cooling age of that section of the gneiss complex whose K–Ar isotopic system was completely reset by the pre-existing metamorphic processes. Ages obtained from dolerite dykes in the area range from 1840 ± 20 Ma to 3160 ± 20 Ma and admit no simple interpretation, nor do they afford evidence to endorse the above interpretation made of the age data. It is perhaps significant that the age of 2410 ± 40 Ma lies close to the oldest age reported by Larson and Møller [14] from the Central basement gneiss unit of West Greenland, close to 2500 Ma on biotite from a hypersthene-biotite gneiss similar to sample G11. Subsequent isotopic investigations of the Archaean block have concentrated on

TABLE 3

K–Ar ages from Central West Greenland

		K_2O (%)	^{40}Ar radiogenic ($mm^3\ g^{-1}$)	Age (Ma)	Age corresponding to best linear fit		Inherited ^{40}Ar ($\times 10^{-1}\ mm^3\ g^{-1}$)
Itivdleq area							
Basic	G14/6W	0.89	0.0706	1549	1700 ± 40		−0.074 ± 0.037
Gneiss	G14/6L	0.50	0.0402	1560			
Band	G14/6H	2.50	0.220	1650			
Acid	G18/11W	2.55	0.320	2080	1830 ± 20		0.594 ± 0.016
Gneiss	G18/11L	0.55	0.117	2810			
Band	G18/11H	8.47	0.930	1914			
Dolerite	D14/7	0.64	0.0644	1800 ± 20			
Dyke	D14/9	0.63	0.0578	1720 ± 20			
Dolerite Dyke	D17/5	0.73	0.0795	1890 ± 20			
Craton							
Acid	G11/6W	0.89	0.123	2200	2420 ± 30		
Gneiss	G11/6L	0.35	0.0345	1780			
Band	G11/6H	1.25	0.181	2260			
	G11/7W	0.94	0.137	2270	2060 ± 30	2410 ± 40	
	G11/7L	0.73	0.111	2325			−0.166 ± 0.053
	G11/9L	0.55	0.0738	2160	2400 ± 30		
	G11/9H	1.67	0.254	2330			
Dolerite	D3/4	0.57	0.156	3160 ± 40			
Dykes	D4/1	0.43	0.0635	2305 ± 30			
	D6/2	0.90	0.106	2010 ± 20			
	D6/7	0.88	0.0911	1840 ± 20			
Sagdlerssuaq							
Basic	G5W	1.15	0.112	1770			
Gneiss	G5L	0.48	0.0603	2075	1660 ± 80		0.159 ± 0.085
Bands	G5H	1.70	0.168	1790			
	G5/6W	1.12	0.115	1835			
	G5/6L	0.45	0.0622	2200	1640 ± 70		0.221 ± 0.072
	G5/6H	1.44	0.149	1845			
	G6W	1.88	0.164	1635	1660 ± 30		−0.030 ± 0.020
	G6L	0.19	0.0148	1528			
	G7W	1.18	0.114	1765			
	G7L	0.37	0.0474	2105	1590 ± 20		0.169 ± 0.002
	G7H	1.53	0.143	1725			
	G8W	1.16	0.144	2065			
	G8L	0.13	0.0575	4800	1580 ± 30		0.476 ± 0.019
	G8H	2.07	0.216	1855			
			Mean of 4 concordant ages		1620 ± 50		

$\lambda_e = 0.584 \times 10^{-10}\ yr^{-1}$; $\lambda_\beta = 4.72 \times 10^{-10}\ yr^{-1}$; $^{40}K/K = 1.19 \times 10^{-2}$ atom percent.

Rb–Sr and U–Pb methods and further corroborative K–Ar data, possibly to define a regional cooling age, is regrettably lacking.

4. Conclusions

1 It has recently been pointed out [20,21,12] that in terrains which have cooled over many millions of years any stable magnetization would have been imposed well below the Curie point, perhaps as low as 200°C and that as a consequence the blocking temperatures for the magnetic minerals and for the K–Ar mineral systems would probably be similar. (But at such temperatures cooling rates would be very slow so that even a small difference in blocking temperatures might result in a significant time difference.) In the present case, the radiometric and palaeomagnetic record would have been fixed during cooling and uplift after the Nagssugtoqidian metamorphism had ceased.

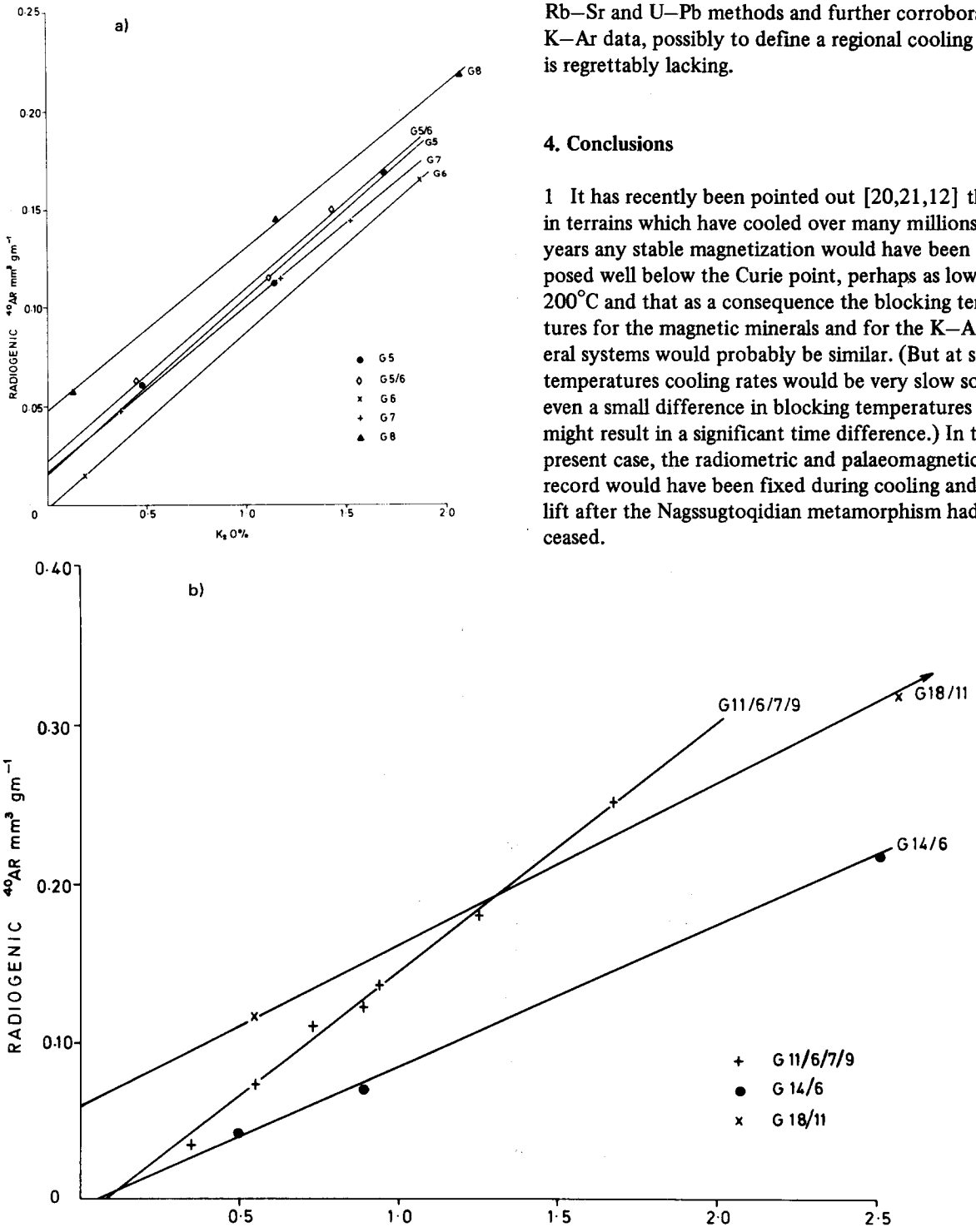


Fig. 7. K–Ar data of gneisses from Central West Greenland: (a) Sagdlerssuaq, (b) Itivdleq and the Archaean craton.

It has been seen that the Sagdlerssuaq rocks record a later cooling age than those at Itivdleq. A likely explanation of this is that brittle overthrusting from the NNW [2] implies that the level exposed at Sagdlerssuaq was originally deeper than that now seen at Itivdleq. Moreover, the magnetization at Sagdlerssuaq would also be younger than that observed at Itivdleq by an amount related to the difference of their radiometric ages. We propose that the direction of magnetization from the Itivdleq area was established at a time close to 1750 Ma, although the lack of agreement between the measured isochron ages prohibits a precise specification of this age. At Sagdlerssuaq the corresponding age was determined to be 1620 (±50) Ma.

A direction of magnetization (declination = 277°, inclination = 69°, α_{95} = 14°) indistinguishable from that of Sagdlerssuaq has been determined by Fahrig and Bridgwater [11] from amphibolite boudins derived from Kangâmiut dykes "north of the Nagssugtoqidian boundary". These authors share our view that this direction of magnetization was imposed after that at Itivdleq. However, we differ from them in that we believe that the Itivdleq dykes were magnetized during regional cooling and not at the time of intrusion; this view is endorsed by Morgan [12].

A palaeomagnetic pole similar to that from Sagdlerssuaq has also been determined by J.D.A. Piper (personal communication) for pillow lavas and supracrustals from Arsuk in the Ketilidian of southwest Greenland. There, magnetization occurred after regional folding and was imposed between 1600 and 1800 Ma.

These poles are unlike poles of similar ages determined from the Canadian Shield. Piper has suggested [22] that the apparent polar wander path for North America is more complicated than has yet been recognized. On the other hand the Itivdleq dyke pole fits the path of Irving and McGlynn [23] rather well, although the fit is not quite so good upon closure of the Davis Strait.

We believe that the work described in this paper defines a portion of the Precambrian polar wander path for Greenland. It also emphasizes the potential values of palaeomagnetism (especially when combined with radiometric methods which are sensitive to relatively low temperatures, such as the K–Ar method) in problems of crustal uplift. The likelihood [12] that palaeomagnetic results from slowly cooled terrains will be free of the effects of secular variation implies that palaeomagnetism is a more precise tool for Precambrian research than has yet been fully appreciated.

Acknowledgements

The rocks discussed in this paper were collected with Dr. G.E. Morgan and formed part of a large collection made by the Geophysics Department, Imperial College, London, under the auspices of Professor J. Sutton, F.R.S. Field support was very generously provided by Dr. Juan Watterson of Liverpool University. After the expedition the rocks were shared by Imperial College and Newcastle University. The palaeomagnetic work of this paper has been carried out (by G.E.J.B.) during the tenure of a Leverhulme Research Fellowship. The palaeomagnetic calculations have been performed with a program written by Dr. A. Richardson with improvements by Dr. H.I. Scoins. I acknowledge a discussion with Dr. J.B. O'Donovan.

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